

Historical Timeline

How the field moved from remembering nothing to managed memory

Deliverable 1 · Conversational Memory Intelligence System · Ketan Juikar · June 2026

This is the chain that leads to a memory system. Read it as **approach → the limit it hit → the next approach**. Each step solved the previous problem and then exposed a new one. The point is not history for its own sake. It is to show that the final design is a reasoned answer to a series of real limits, not a pattern copied from someone else's code.

STEP 0 · THE STARTING POINT

Remember nothing

The approach	Every request is independent. The model sees the current message and nothing else.
It assumes	Every question is self-contained, so no past is needed.
The limit	Across many sessions, the user has to set up all the context every single time. Information is lost between conversations, and that is the exact cost the whole field then spends a decade trying to remove.



STEP 1 · THE FIRST FIX

Replay the whole conversation every time

The approach	Paste the entire history into every request. "Just give the model everything."
It assumes	The context window is big enough, and cost and speed do not matter.
The limit	The context window is fixed, and the cost of attention grows with the square of the input length. So long relationships overflow it. Even before that, cost and speed get worse every turn, because the whole history is re-read each time.

Research anchor: Vaswani et al., Attention Is All You Need (2017), and the long-context work that follows it.



STEP 2 · MEMORY AS A SEPARATE THING

Give the model an external memory

The approach	Give the network a memory outside itself that it can read from and write to, learned as part of training.
It assumes	A learnable external memory is the right way to carry information forward.
The limit	These are research designs, not a practical store for real user facts in a real product. They prove that memory should be its own separate component, but they do not answer the practical questions: what deserves to be stored, how to govern it, or how to keep users apart.

Research anchor: Weston et al., Memory Networks (2014); Sukhbaatar et al., End-To-End Memory Networks (2015); Graves et al., Neural Turing Machines (2014).



STEP 3 · SEARCH-BASED MEMORY

Store everything, search by similarity

The approach	Index everything in a vector database, and pull back the closest matches when a question arrives.
It assumes	Similar meaning is a good enough stand-in for relevance, and storing everything is fine because search will filter it.
The limit	With no gate on what gets stored, junk buries the good memories. Pure similarity misses exact names and surfaces old or contradictory facts. There is no sense of recency, no importance, no forgetting, and no separation between users. Storage was solved. Relevance and safety were not.

Research anchor: Lewis et al., Retrieval-Augmented Generation (2020); Borgeaud et al., RETRO (2022); Wu et al., Memorizing Transformers (2022).



STEP 4 · WHERE THIS PROJECT LANDS

Managed memory

The approach	A selective, typed, ranked, fading, governed memory. A gate that decides what gets stored, a proper record, ranking that uses several signals, hybrid search, a nightly forgetting job, and separation enforced by the database.
It assumes	A good memory system is mostly a forgetting system, and relevance is a scoring problem rather than a lookup.
Where it lands	This is the kind of system this project builds. The specifics belong in the system design. What matters here is that every part of it answers one of the limits above.

How to read this chain. Notice that no step was "wrong." Each was the right next move given the limit in front of it, and each one created the conditions that made the next move necessary. That is the whole argument for the final design. It is not the architecture I liked best. It is the one left standing after every simpler option was forced to give way.